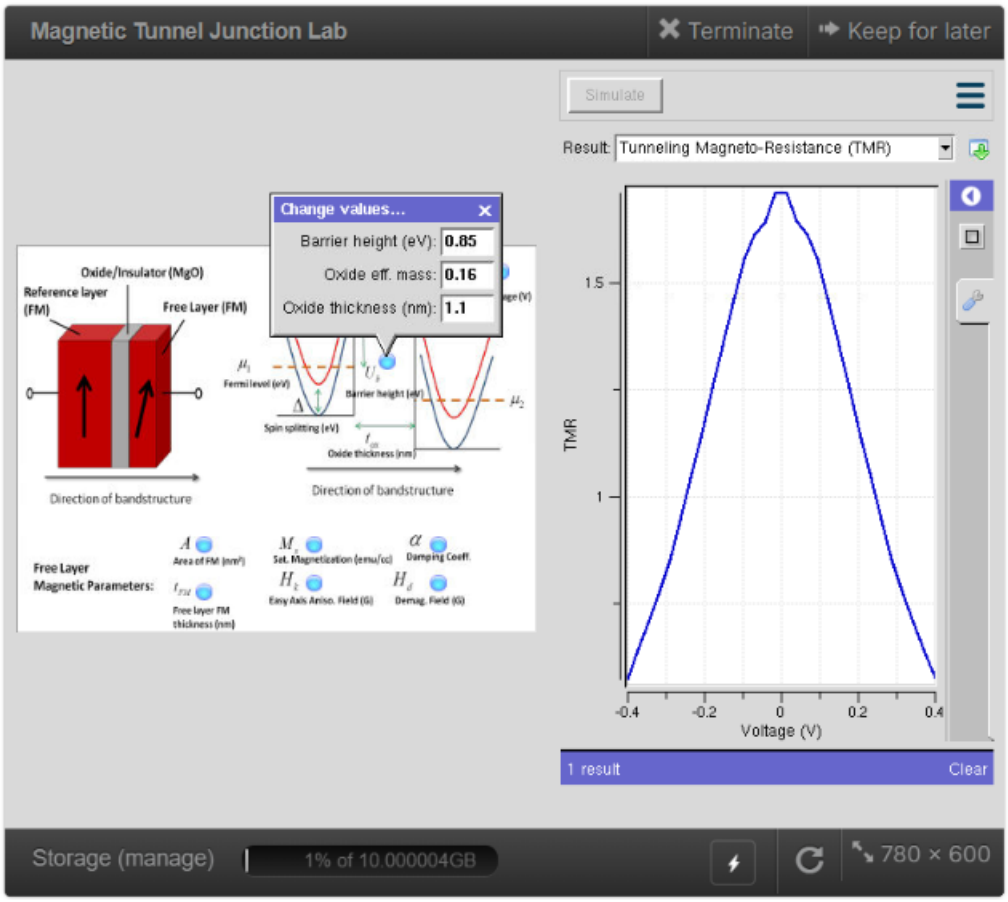
Exp. 9. Simulation of Tunneling Magnetoresistance (TMR) by Varying Barrier Height using NanoHUB

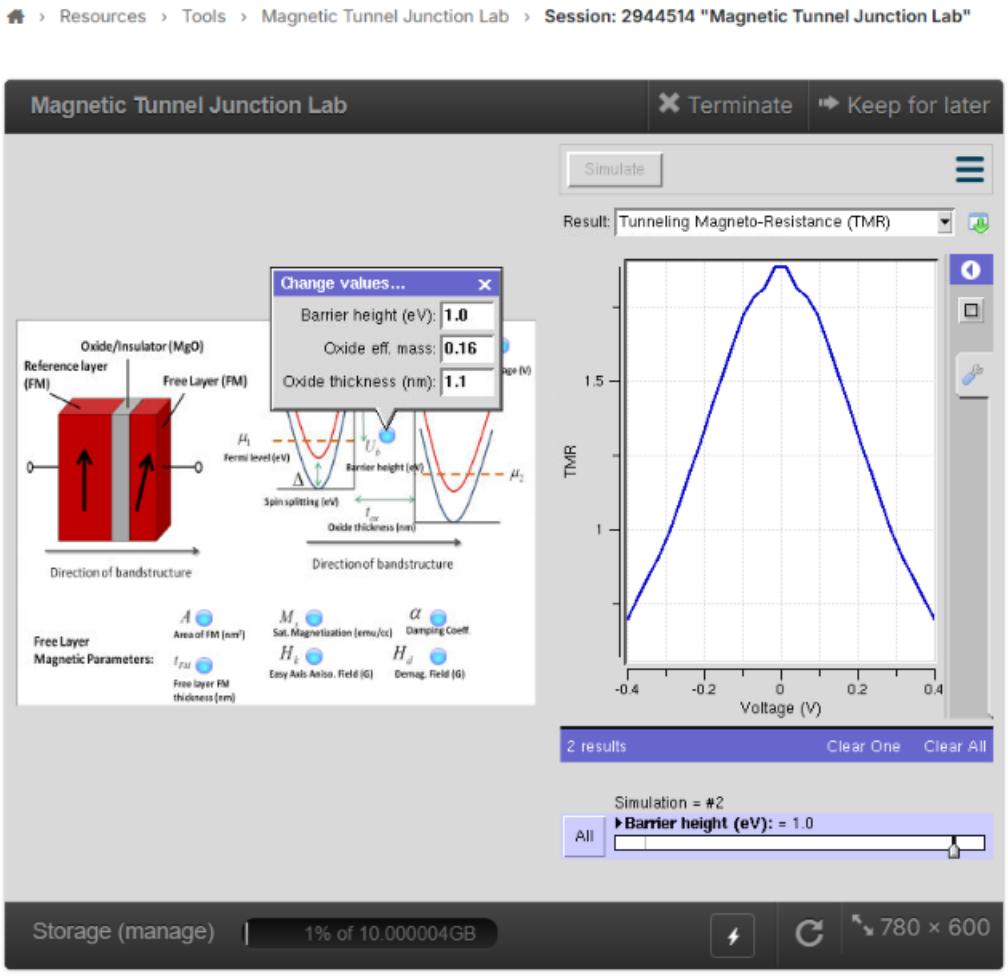
objective:

To obtain the Tunnelling Magnetoresistance (TMR) by varying barrier height. Software: NanoHub Online Tube

Case Study 1: Characteristics of TMR with barrier height of 0.85eV



Case Study 2: Characteristics of TMR with barrier height of 1.0eV



Reflection:

- simulated spin-polarized electron transport across a thin insulating barrier inside a magnetic tunnel junction lab.
- verified that the TMR ratio behaves as a purely quantum mechanical effect dictated by ferromagnet and oxide pairings.
- observed how swept barrier parameters dynamically modulate the magnetoresistance profile over distinct operational voltage biases.